

practice-01

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1 *Indian Liver Patient*

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The given data is collected from the given [link](#)

1.1 *Data Set Description*

1.1.1 Context:

Patients with Liver disease have been continuously increasing because of excessive consumption of alcohol, inhale of harmful gases, intake of contaminated food, pickles and drugs. This dataset was used to evaluate prediction algorithms in an effort to reduce burden on doctors.

1.1.2 Content:

This data set contains 416 liver patient records and 167 non liver patient records collected from North East of Andhra Pradesh, India. The “Dataset” column is a class label used to divide groups into liver patient (liver disease) or not (no disease). This data set contains 441 male patient records and 142 female patient records.

Any patient whose age exceeded 89 is listed as being of age “90”.

1.1.3 Columns:

Age of the patient Gender of the patient Total Bilirubin Direct Bilirubin Alkaline Phosphatase Alamine Aminotransferase Aspartate Aminotransferase Total Protiens Albumin Albumin and Globulin Ratio Dataset: field used to split the data into two sets (patient with liver disease, or no disease) ### **Acknowledgements:** This dataset was downloaded from the UCI ML Repository:

Lichman, M. (2013). UCI Machine Learning Repository [<http://archive.ics.uci.edu/ml>]. Irvine, CA: University of California, School of Information and Computer Science.

1.1.4 Inspiration:

Use these patient records to determine which patients have liver disease and which ones do not.

1.2 Purpose of the data

To check patience of liver and non liver diseaces

```
[5]: # Importing the libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
[7]: df = pd.read_csv('./data/indian_liver_patient.csv')
```

```
[8]: df.head()
```

```
[8]:
```

	Age	Gender	Total_Bilirubin	Direct_Bilirubin	Alkaline_Phosphotase	\
0	65	Female	0.7	0.1	187	
1	62	Male	10.9	5.5	699	
2	62	Male	7.3	4.1	490	
3	58	Male	1.0	0.4	182	
4	72	Male	3.9	2.0	195	

	Alamine_Aminotransferase	Aspartate_Aminotransferase	Total_Protiens	\
0	16	18	6.8	
1	64	100	7.5	
2	60	68	7.0	
3	14	20	6.8	
4	27	59	7.3	

	Albumin	Albumin_and_Globulin_Ratio	Dataset
0	3.3	0.90	1
1	3.2	0.74	1
2	3.3	0.89	1
3	3.4	1.00	1
4	2.4	0.40	1

```
[9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 583 entries, 0 to 582
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    583 non-null    int64
1   Gender                                583 non-null    object
2   Total_Bilirubin                       583 non-null    float64
3   Direct_Bilirubin                      583 non-null    float64
4   Alkaline_Phosphotase                  583 non-null    int64
5   Alamine_Aminotransferase              583 non-null    int64
6   Aspartate_Aminotransferase            583 non-null    int64
7   Total_Protiens                        583 non-null    float64
8   Albumin                              583 non-null    float64
```

```

    9   Albumin_and_Globulin_Ratio  579 non-null    float64
   10   Dataset                    583 non-null    int64
dtypes: float64(5), int64(5), object(1)
memory usage: 50.2+ KB

```

```
[10]: df.isnull().sum()
```

```

[10]: Age                0
      Gender              0
      Total_Bilirubin     0
      Direct_Bilirubin    0
      Alkaline_Phosphotase 0
      Alamine_Aminotransferase 0
      Aspartate_Aminotransferase 0
      Total_Protiens      0
      Albumin             0
      Albumin_and_Globulin_Ratio 4
      Dataset             0
      dtype: int64

```

```
[21]: df.isnull().sum()/len(df)*100
```

```

[21]: Age                0.000000
      Gender              0.000000
      Total_Bilirubin     0.000000
      Direct_Bilirubin    0.000000
      Alkaline_Phosphotase 0.000000
      Alamine_Aminotransferase 0.000000
      Aspartate_Aminotransferase 0.000000
      Total_Protiens      0.000000
      Albumin             0.000000
      Albumin_and_Globulin_Ratio 0.686106
      Dataset             0.000000
      dtype: float64

```

```
[20]: df.describe().T
```

```

[20]:
      count      mean      std   min   25%  \
Age      583.0    44.746141  16.189833   4.0  33.0
Total_Bilirubin  583.0    3.298799   6.209522   0.4   0.8
Direct_Bilirubin  583.0    1.486106   2.808498   0.1   0.2
Alkaline_Phosphotase  583.0  290.576329  242.937989  63.0  175.5
Alamine_Aminotransferase  583.0   80.713551  182.620356  10.0   23.0
Aspartate_Aminotransferase  583.0  109.910806  288.918529  10.0   25.0
Total_Protiens      583.0    6.483190   1.085451   2.7    5.8
Albumin             583.0    3.141852   0.795519   0.9    2.6
Albumin_and_Globulin_Ratio  579.0    0.947064   0.319592   0.3    0.7

```

Dataset	583.0	1.286449	0.452490	1.0	1.0
	50%	75%	max		
Age	45.00	58.0	90.0		
Total_Bilirubin	1.00	2.6	75.0		
Direct_Bilirubin	0.30	1.3	19.7		
Alkaline_Phosphotase	208.00	298.0	2110.0		
Alamine_Aminotransferase	35.00	60.5	2000.0		
Aspartate_Aminotransferase	42.00	87.0	4929.0		
Total_Protiens	6.60	7.2	9.6		
Albumin	3.10	3.8	5.5		
Albumin_and_Globulin_Ratio	0.93	1.1	2.8		
Dataset	1.00	2.0	2.0		

```
[24]: df['Albumin_and_Globulin_Ratio'].fillna(df['Albumin_and_Globulin_Ratio'].mean(),
↪, inplace= True )
```

```
[25]: df.isnull().sum()
```

```
[25]: Age                0
      Gender             0
      Total_Bilirubin    0
      Direct_Bilirubin   0
      Alkaline_Phosphotase 0
      Alamine_Aminotransferase 0
      Aspartate_Aminotransferase 0
      Total_Protiens     0
      Albumin            0
      Albumin_and_Globulin_Ratio 0
      Dataset            0
      dtype: int64
```

```
[26]: df.columns
```

```
[26]: Index(['Age', 'Gender', 'Total_Bilirubin', 'Direct_Bilirubin',
           'Alkaline_Phosphotase', 'Alamine_Aminotransferase',
           'Aspartate_Aminotransferase', 'Total_Protiens', 'Albumin',
           'Albumin_and_Globulin_Ratio', 'Dataset'],
          dtype='object')
```

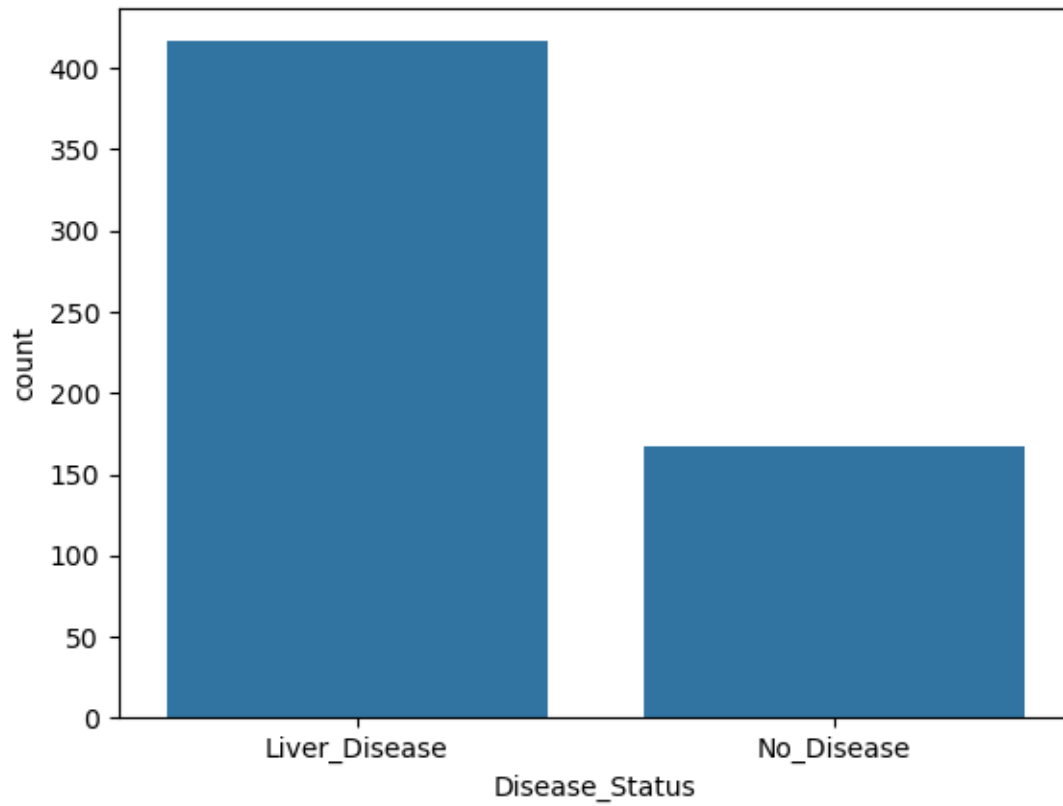
```
[29]: df['Dataset'].value_counts()
```

```
[29]: Dataset
1      416
2      167
      Name: count, dtype: int64
```

```
[30]: df['Disease_Status'] = df['Dataset'].map({1: 'Liver_Disease', 2: 'No_Disease'})
```

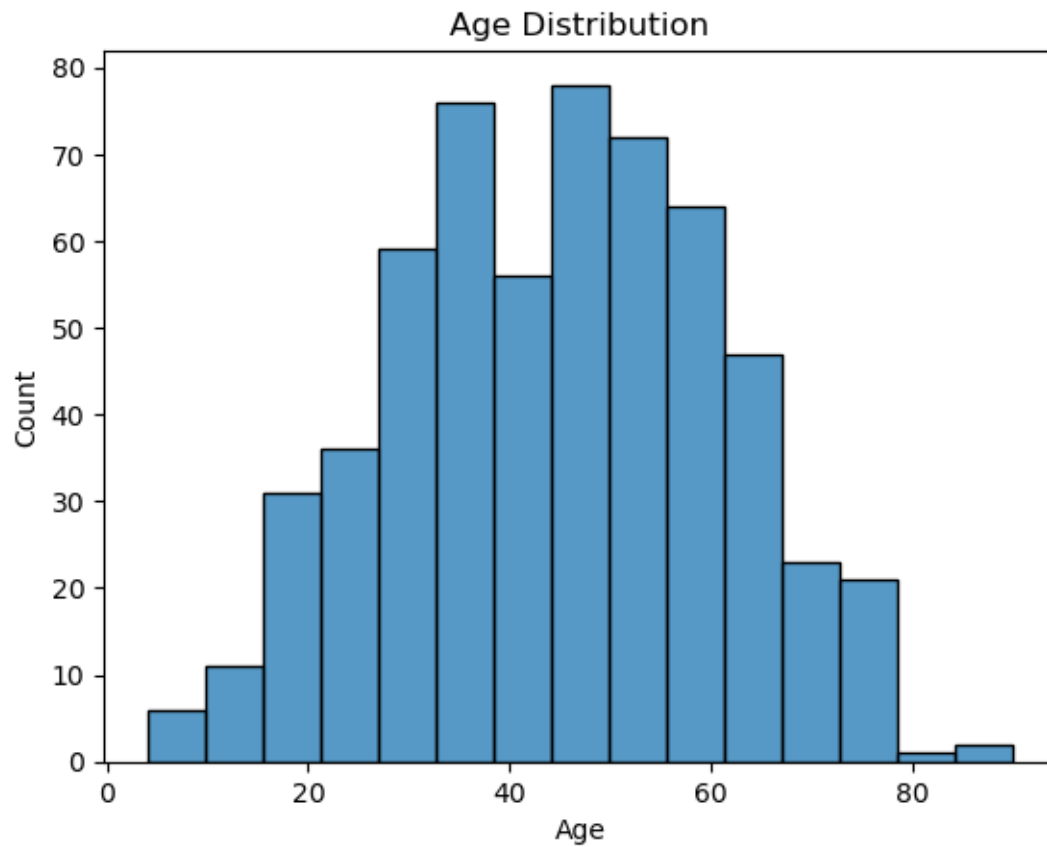
```
[98]: sns.countplot(df, x='Disease_Status')
```

```
[98]: <Axes: xlabel='Disease_Status', ylabel='count'>
```



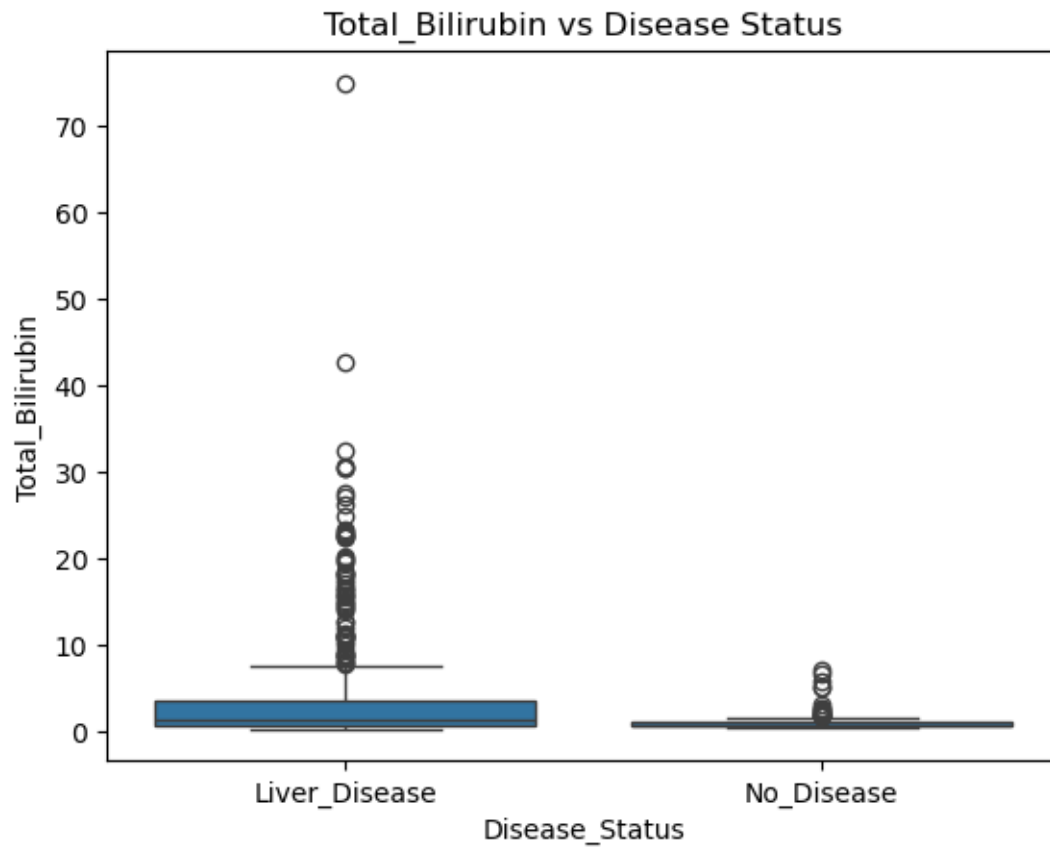
```
[97]: sns.histplot(df['Age'])  
plt.title('Age Distribution')
```

```
[97]: Text(0.5, 1.0, 'Age Distribution')
```



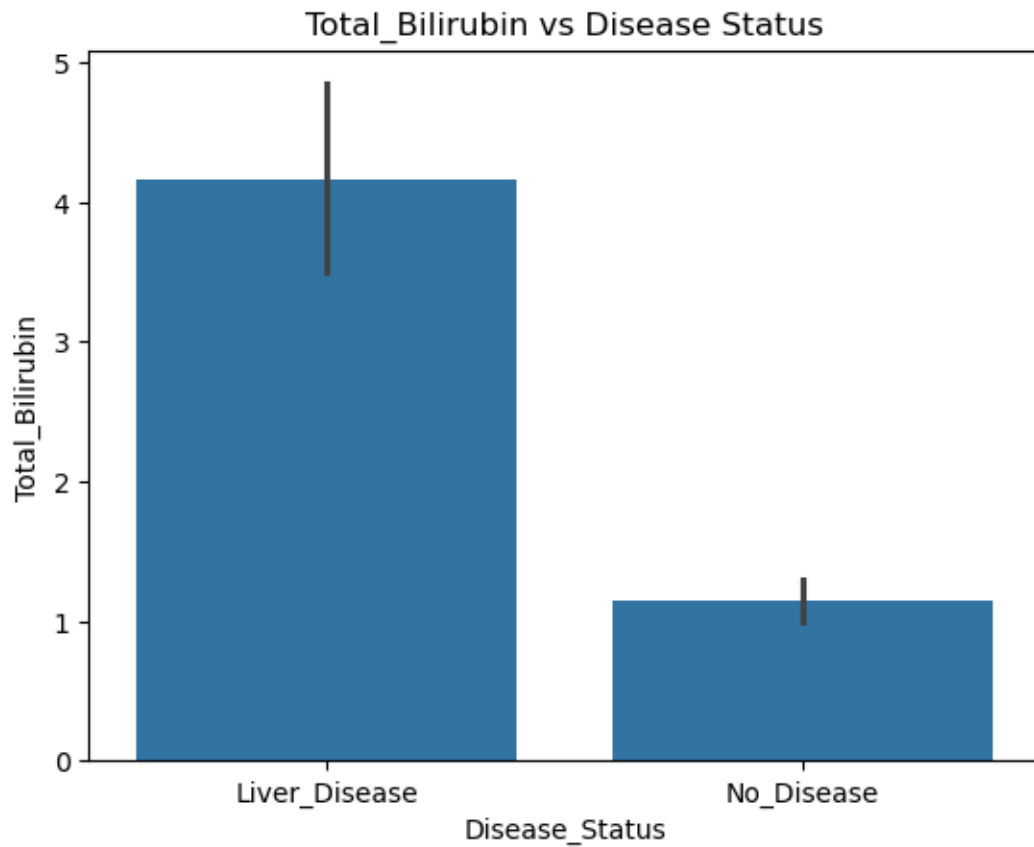
```
[51]: sns.boxplot(df , x='Disease_Status' , y='Total_Bilirubin')  
plt.title('Total_Bilirubin vs Disease Status')
```

```
[51]: Text(0.5, 1.0, 'Total_Bilirubin vs Disease Status')
```



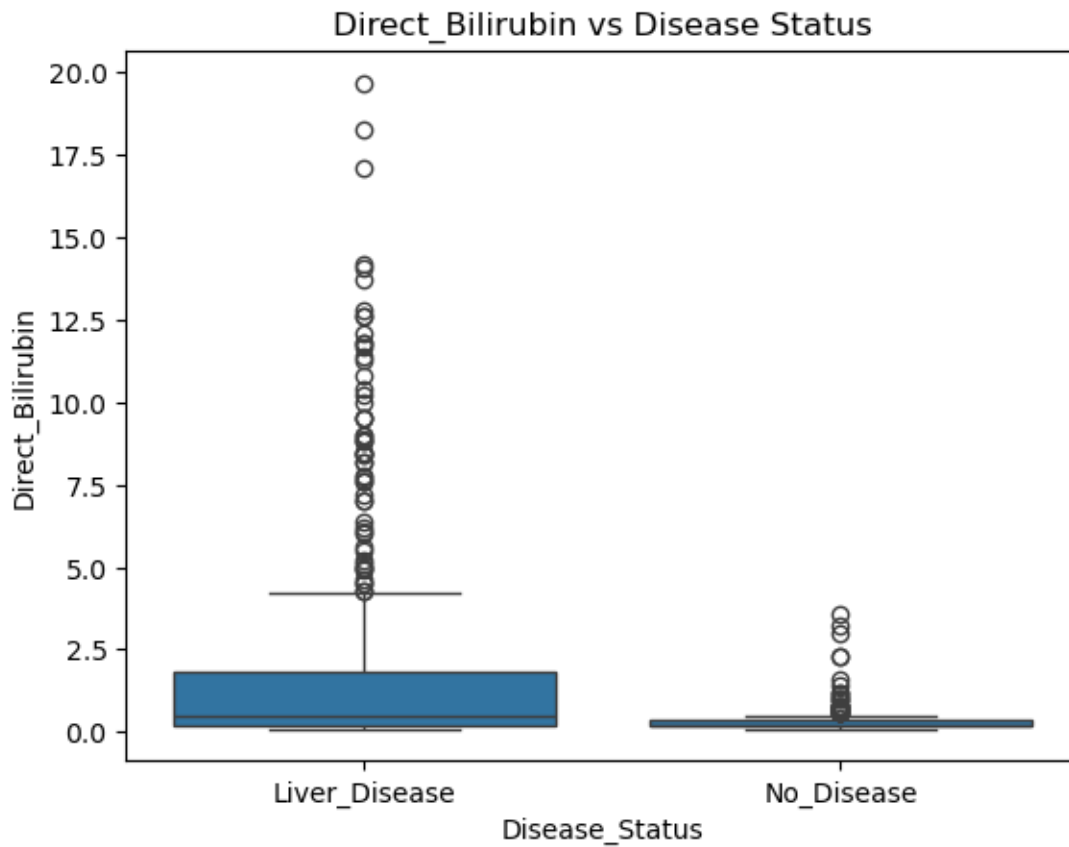
```
[89]: sns.barplot(df , x='Disease_Status' , y='Total_Bilirubin')  
plt.title('Total_Bilirubin vs Disease Status')
```

```
[89]: Text(0.5, 1.0, 'Total_Bilirubin vs Disease Status')
```



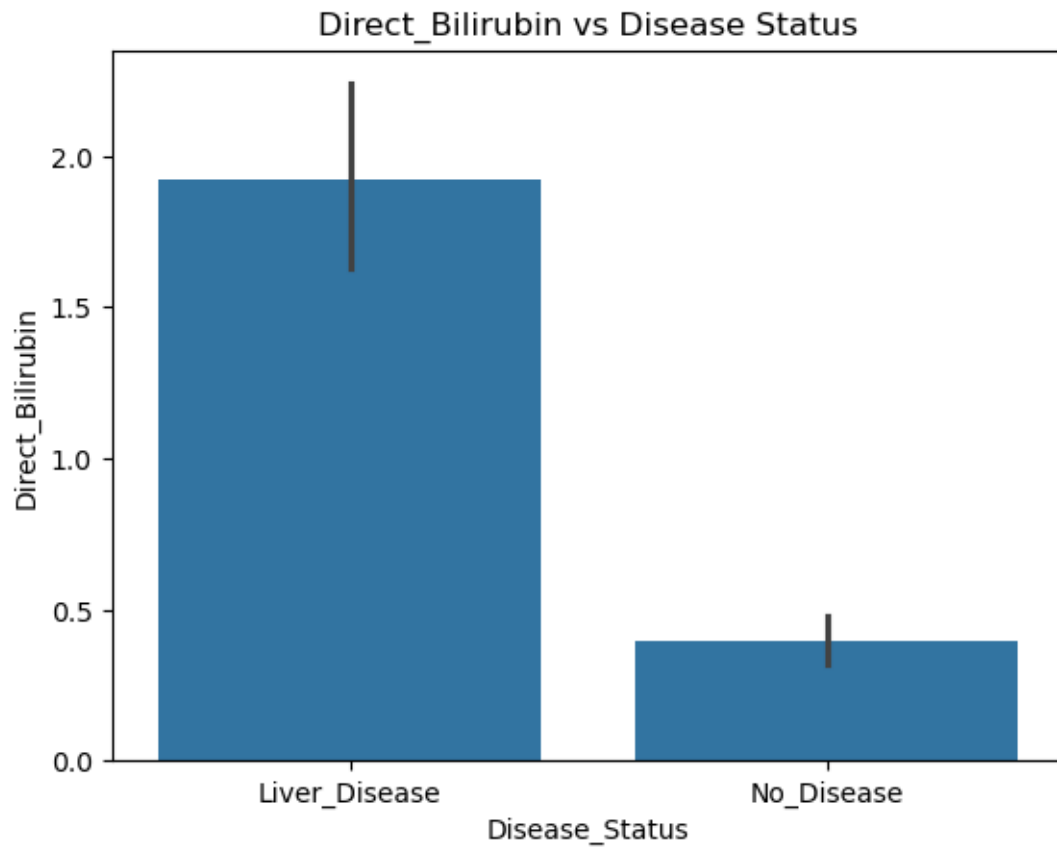
```
[82]: sns.boxplot(df , x='Disease_Status' , y= 'Direct_Bilirubin')  
plt.title('Direct_Bilirubin vs Disease Status')
```

```
[82]: Text(0.5, 1.0, 'Direct_Bilirubin vs Disease Status')
```



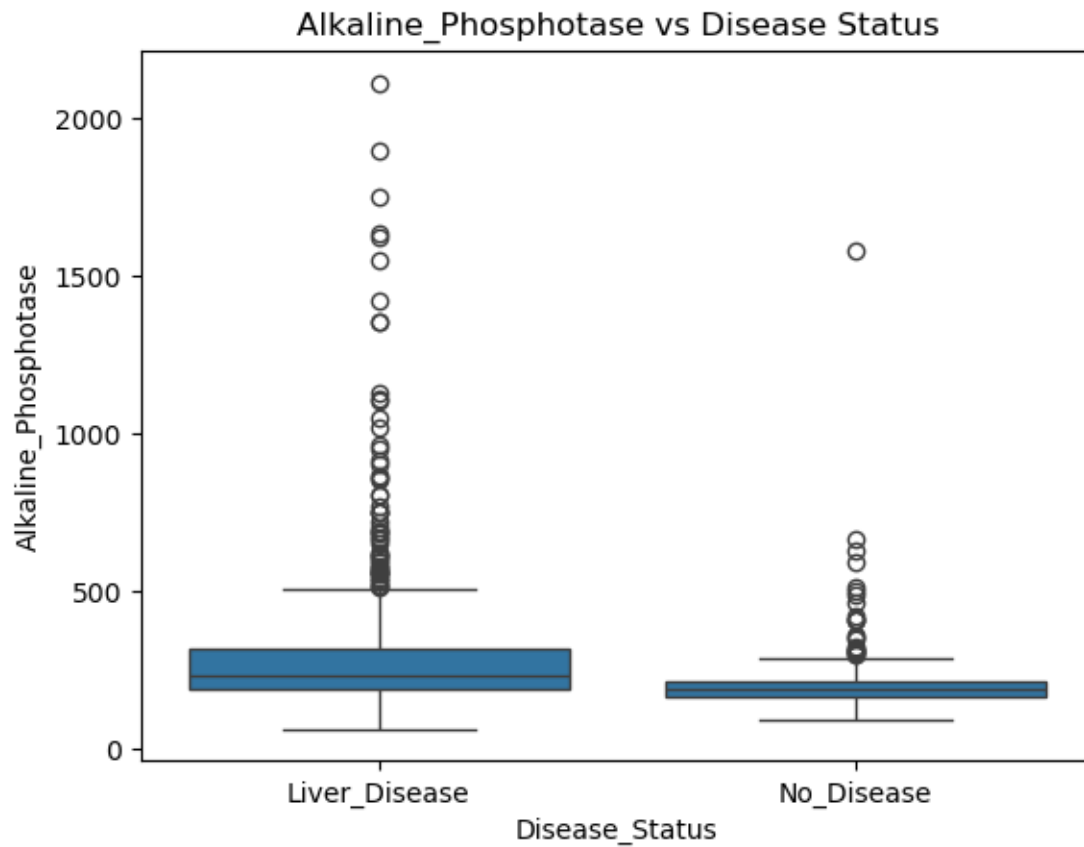
```
[88]: sns.barplot(df , x='Disease_Status' , y= 'Direct_Bilirubin')  
plt.title('Direct_Bilirubin vs Disease Status')
```

```
[88]: Text(0.5, 1.0, 'Direct_Bilirubin vs Disease Status')
```



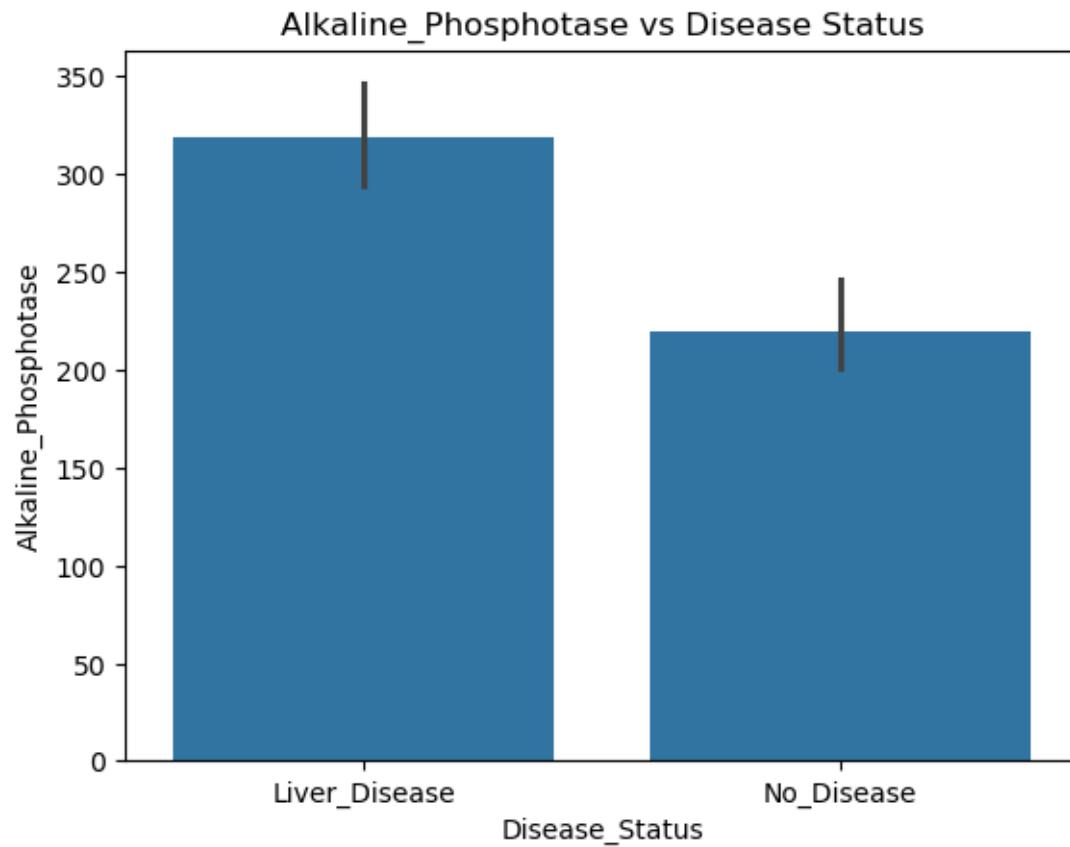
```
[83]: sns.boxplot(df , x='Disease_Status' , y='Alkaline_Phosphotase')  
plt.title('Alkaline_Phosphotase vs Disease Status')
```

```
[83]: Text(0.5, 1.0, 'Alkaline_Phosphotase vs Disease Status')
```



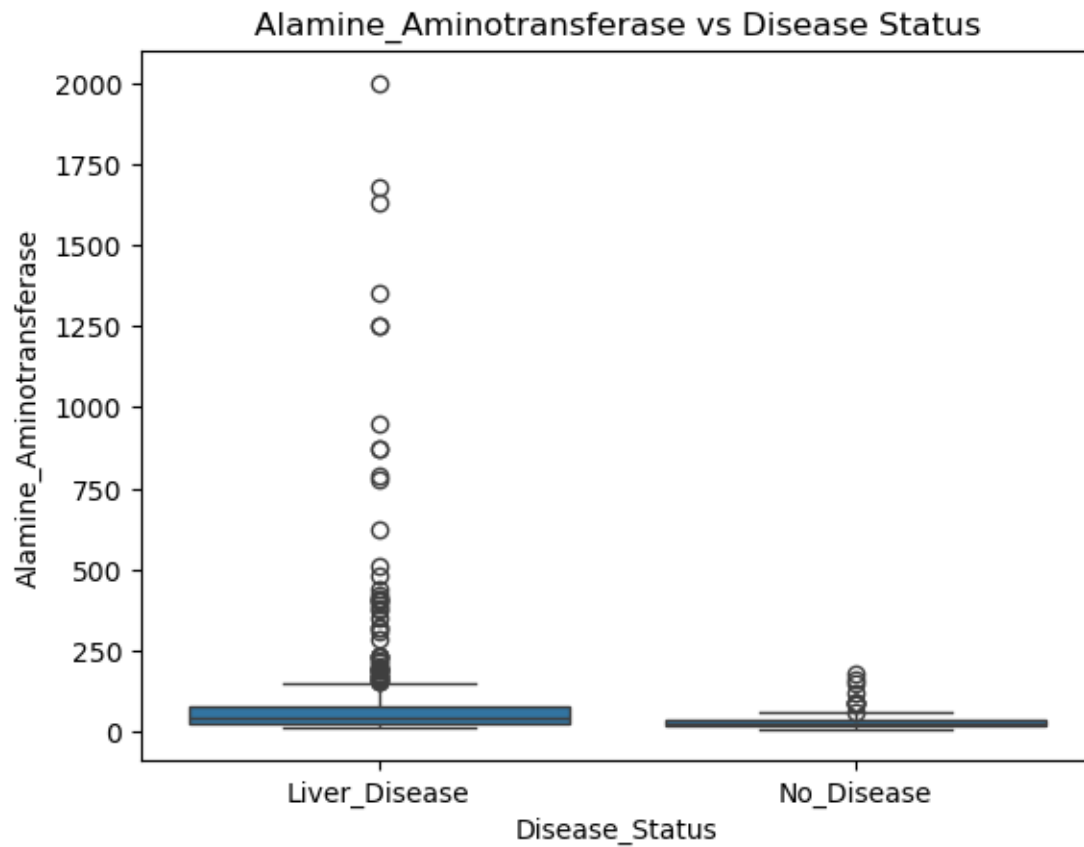
```
[86]: sns.barplot(df , x='Disease_Status' , y='Alkaline_Phosphotase')  
plt.title('Alkaline_Phosphotase vs Disease Status')
```

```
[86]: Text(0.5, 1.0, 'Alkaline_Phosphotase vs Disease Status')
```



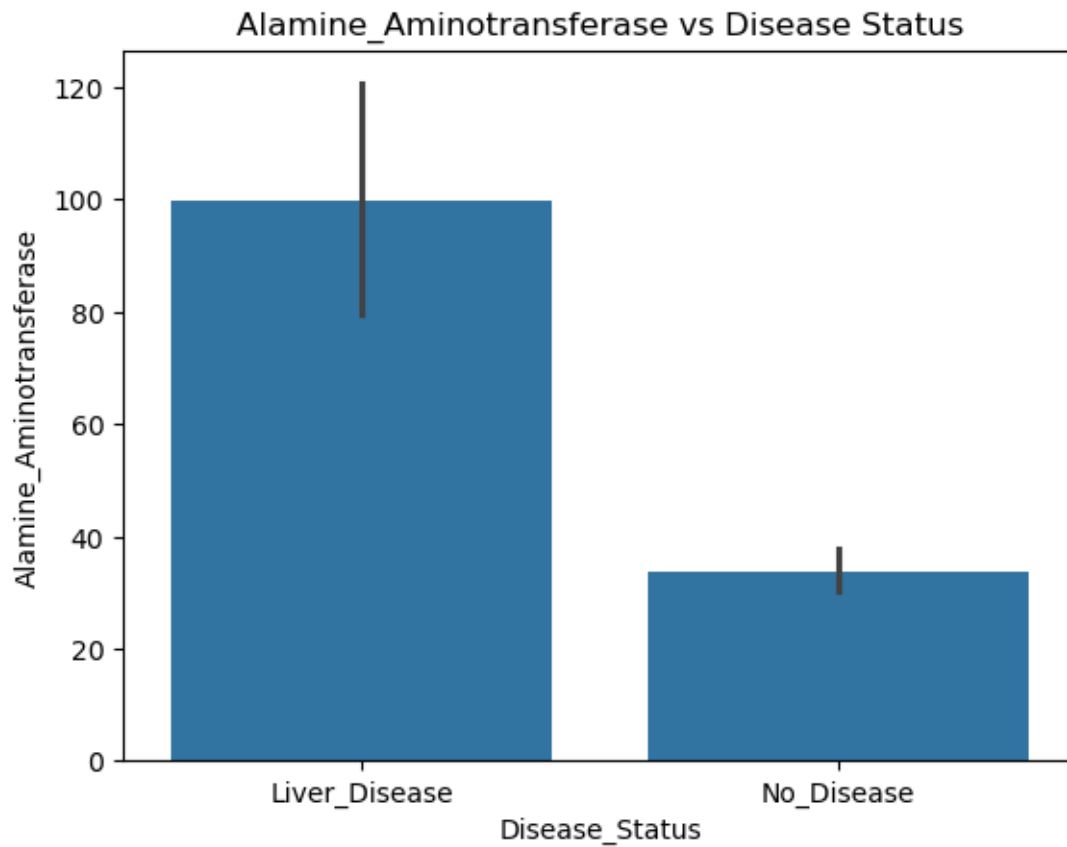
```
[84]: sns.boxplot(df , x='Disease_Status', y= 'Alamine_Aminotransferase')  
plt.title('Alamine_Aminotransferase vs Disease Status')
```

```
[84]: Text(0.5, 1.0, 'Alamine_Aminotransferase vs Disease Status')
```



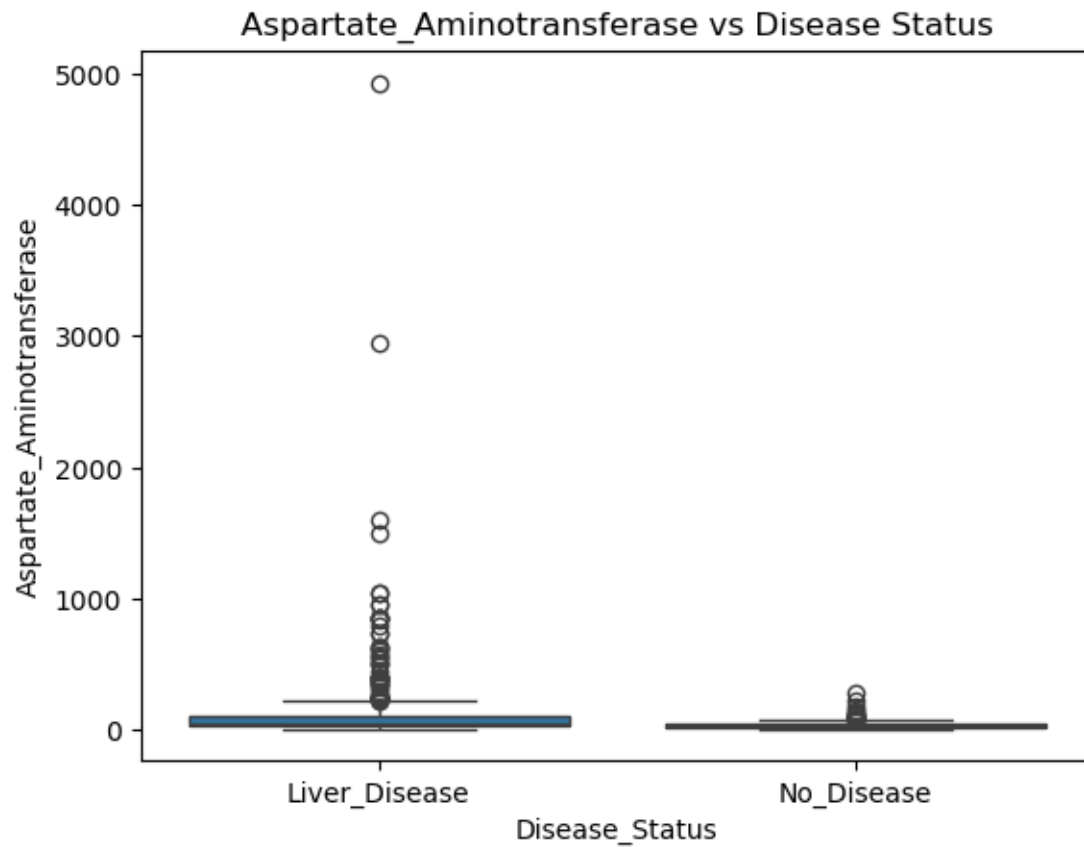
```
[85]: sns.barplot(df , x='Disease_Status', y= 'Alamine_Aminotransferase')
plt.title('Alamine_Aminotransferase vs Disease Status')
```

```
[85]: Text(0.5, 1.0, 'Alamine_Aminotransferase vs Disease Status')
```



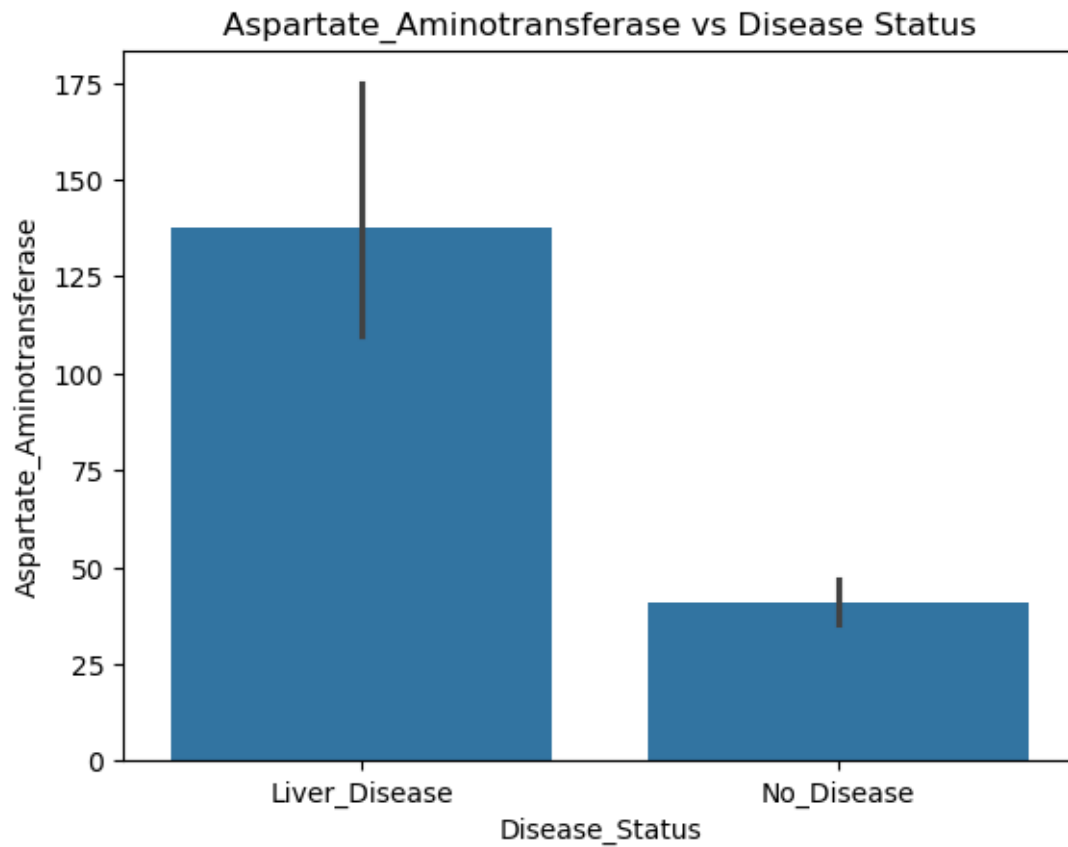
```
[90]: sns.boxplot(df , x='Disease_Status', y= 'Aspartate_Aminotransferase')  
plt.title('Aspartate_Aminotransferase vs Disease Status')
```

```
[90]: Text(0.5, 1.0, 'Aspartate_Aminotransferase vs Disease Status')
```



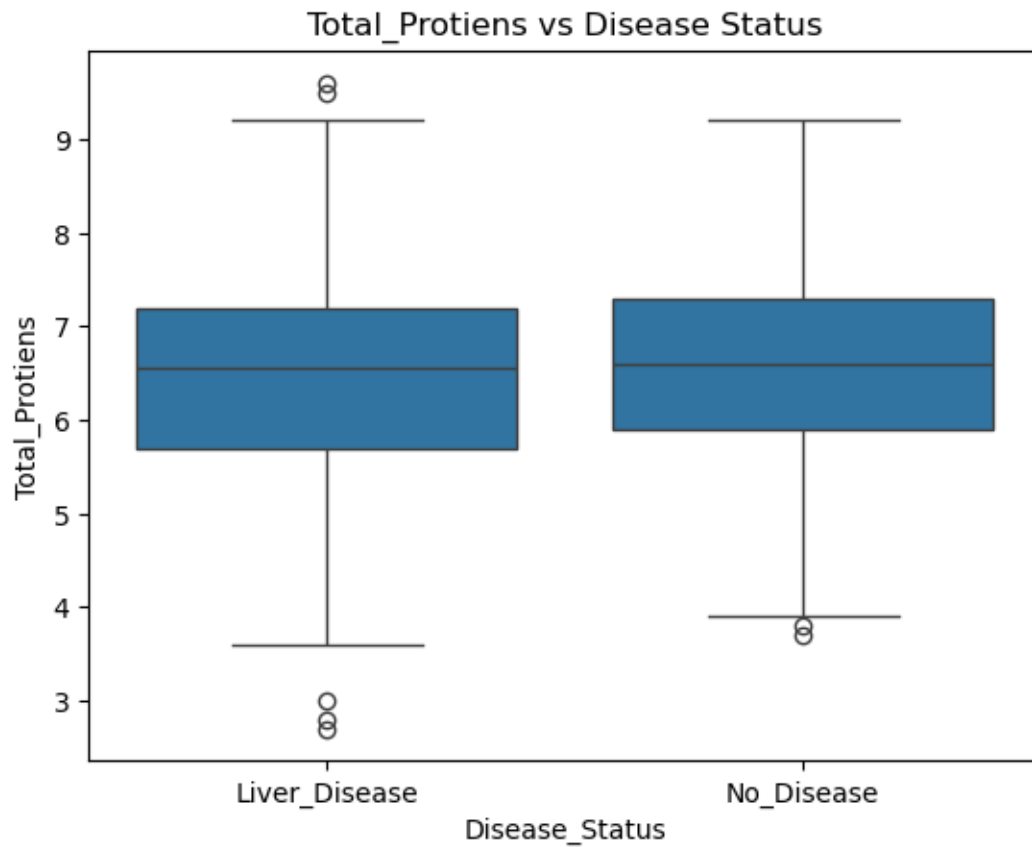
```
[66]: sns.barplot(df , x='Disease_Status', y= 'Aspartate_Aminotransferase')  
plt.title('Aspartate_Aminotransferase vs Disease Status')
```

```
[66]: Text(0.5, 1.0, 'Aspartate_Aminotransferase vs Disease Status')
```



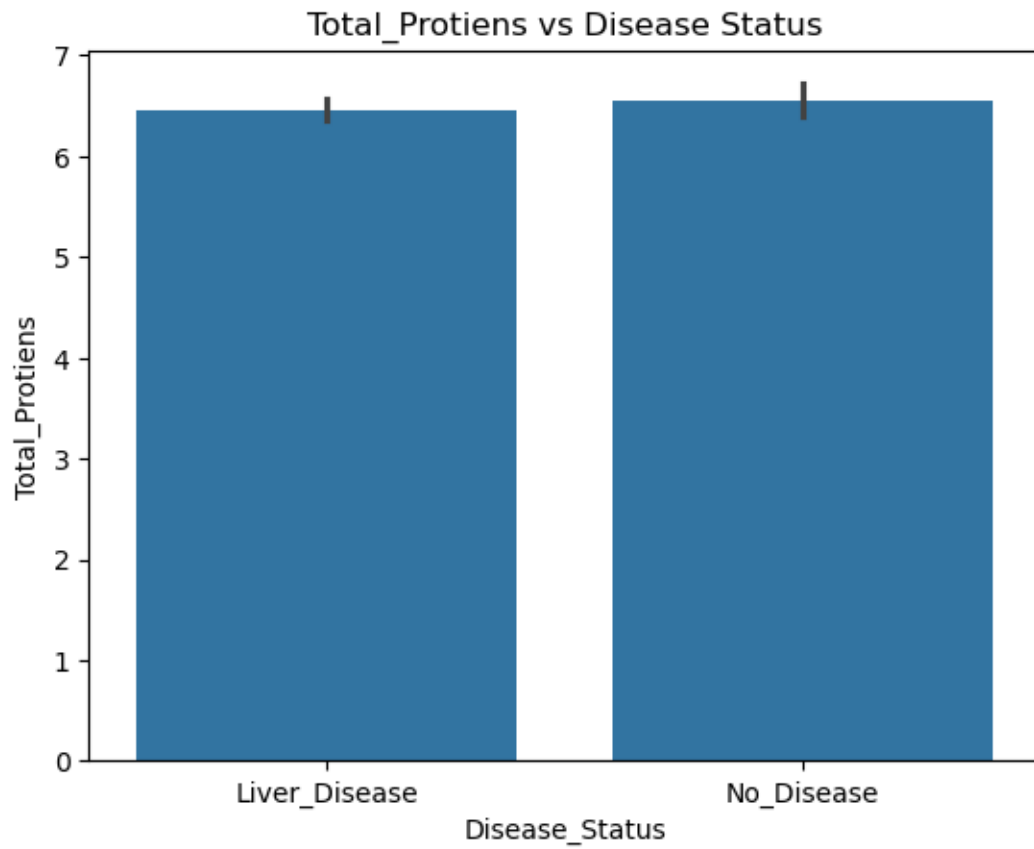
```
[91]: sns.boxplot(df , x='Disease_Status', y='Total_Protiens' )  
plt.title('Total_Protiens vs Disease Status')
```

```
[91]: Text(0.5, 1.0, 'Total_Protiens vs Disease Status')
```



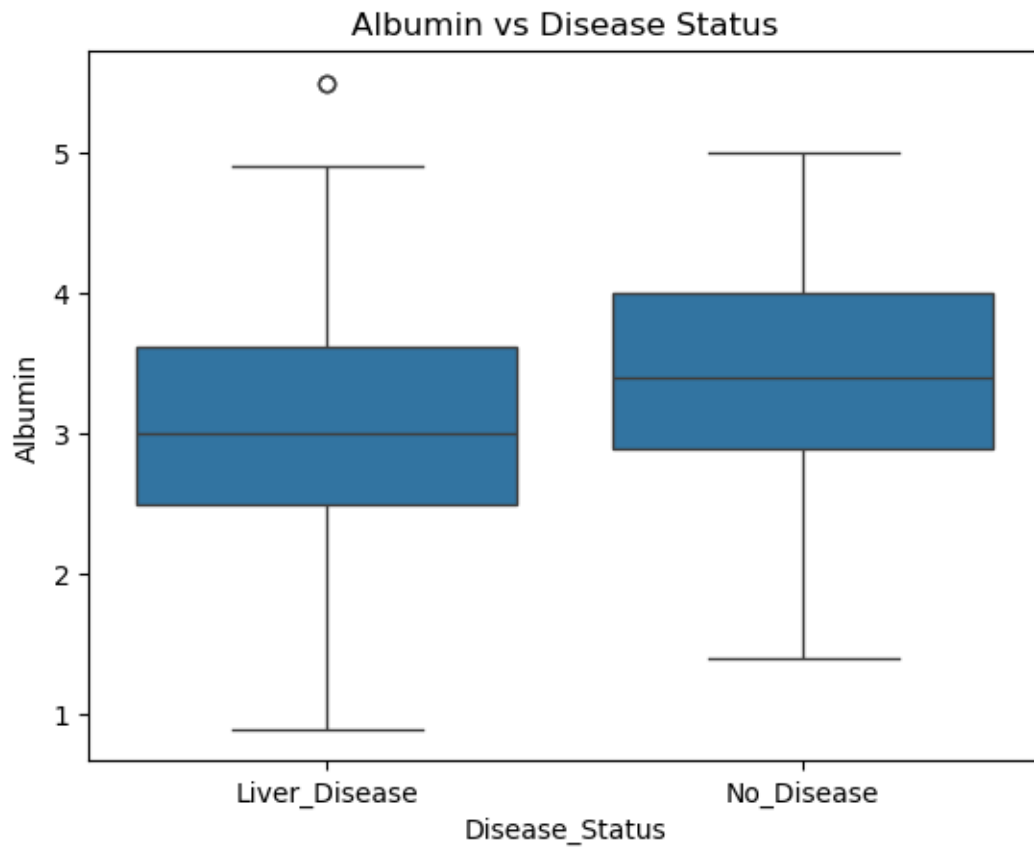
```
[67]: sns.barplot(df , x='Disease_Status', y='Total_Protiens' )  
plt.title('Total_Protiens vs Disease Status')
```

```
[67]: Text(0.5, 1.0, 'Total_Protiens vs Disease Status')
```



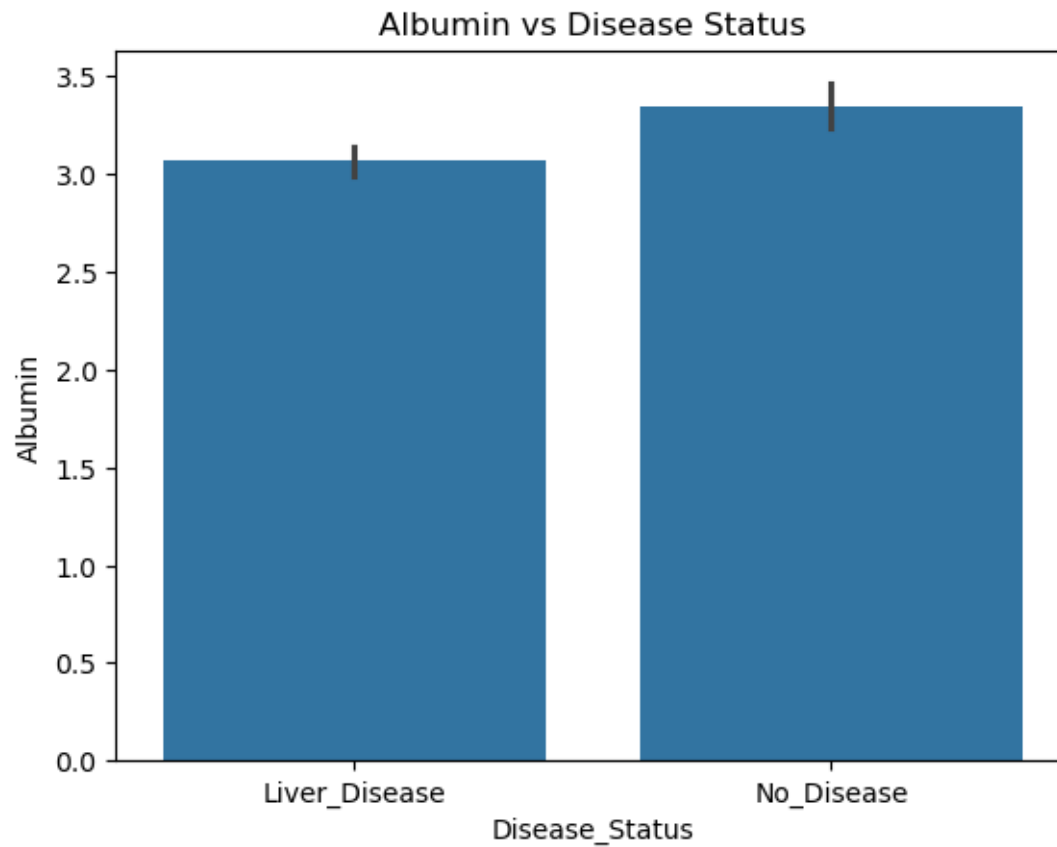
```
[92]: sns.boxplot(df , x='Disease_Status', y='Albumin' )  
plt.title('Albumin vs Disease Status')
```

```
[92]: Text(0.5, 1.0, 'Albumin vs Disease Status')
```



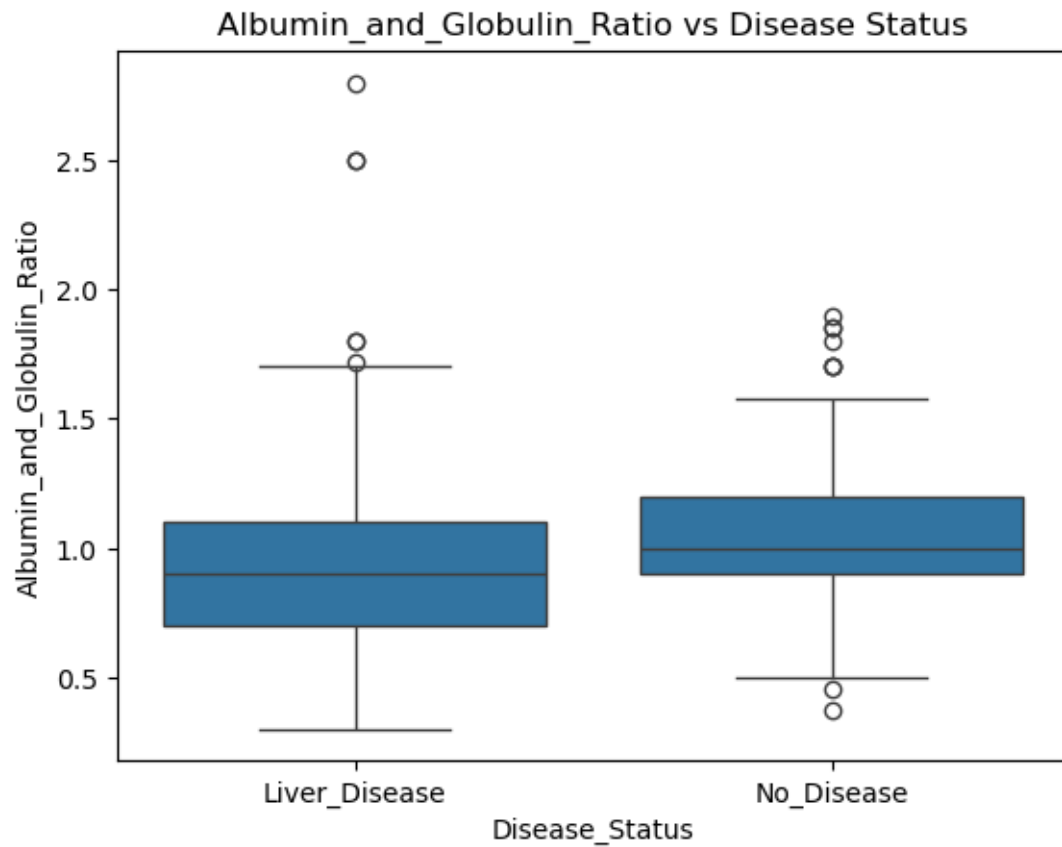
```
[68]: sns.boxplot(df , x='Disease_Status', y='Albumin' )  
plt.title('Albumin vs Disease Status')
```

```
[68]: Text(0.5, 1.0, 'Albumin vs Disease Status')
```



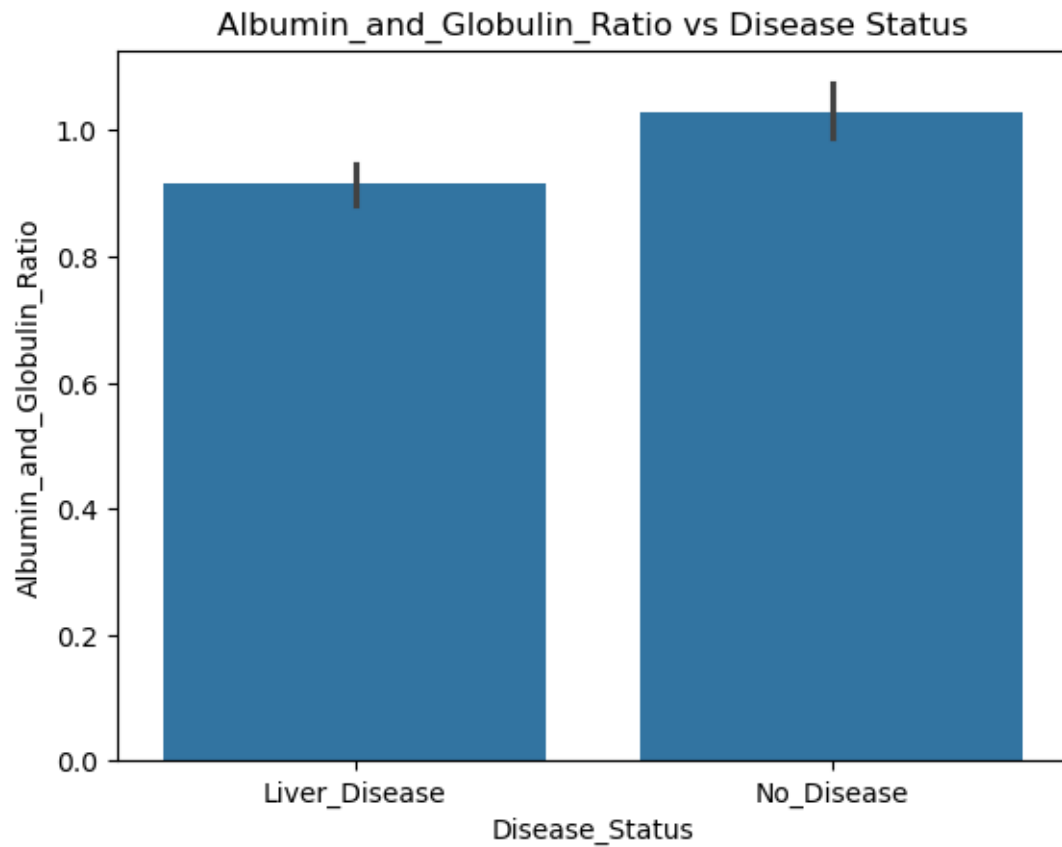
```
[93]: sns.boxplot(df , x='Disease_Status' , y='Albumin_and_Globulin_Ratio')  
plt.title('Albumin_and_Globulin_Ratio vs Disease Status')
```

```
[93]: Text(0.5, 1.0, 'Albumin_and_Globulin_Ratio vs Disease Status')
```



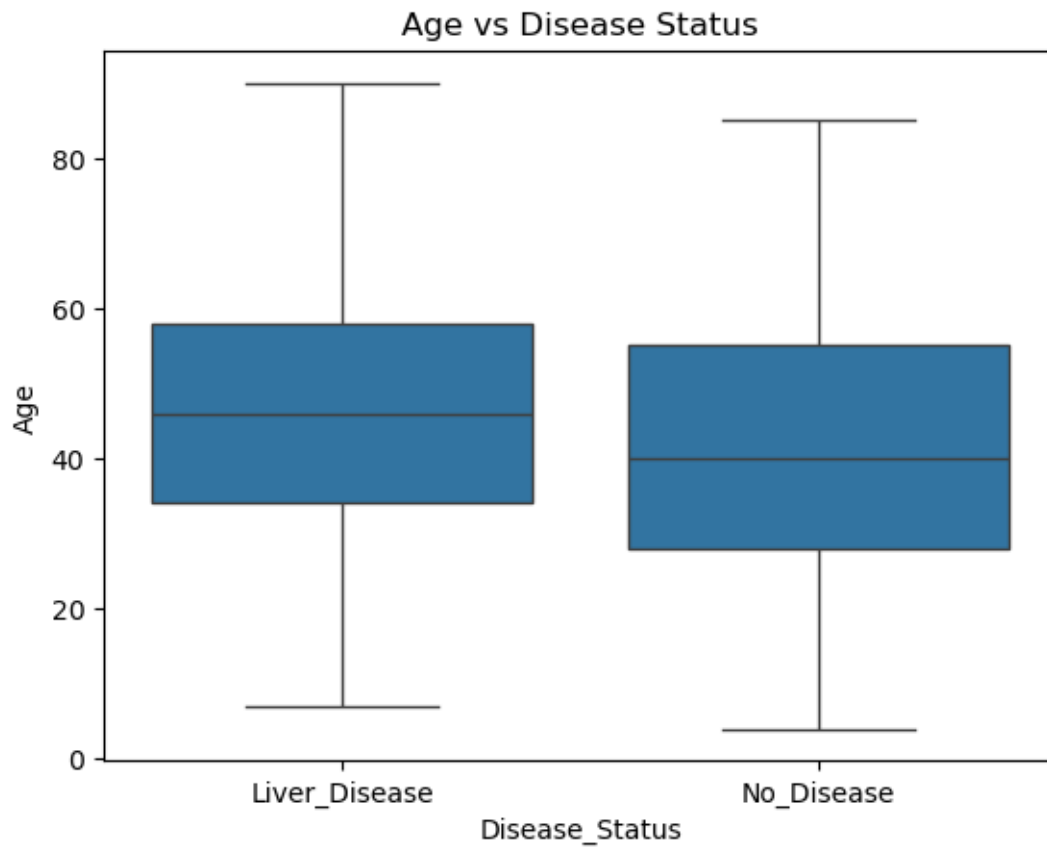
```
[70]: sns.barplot(df , x='Disease_Status' , y='Albumin_and_Globulin_Ratio')  
plt.title('Albumin_and_Globulin_Ratio vs Disease Status')
```

```
[70]: Text(0.5, 1.0, 'Albumin_and_Globulin_Ratio vs Disease Status')
```



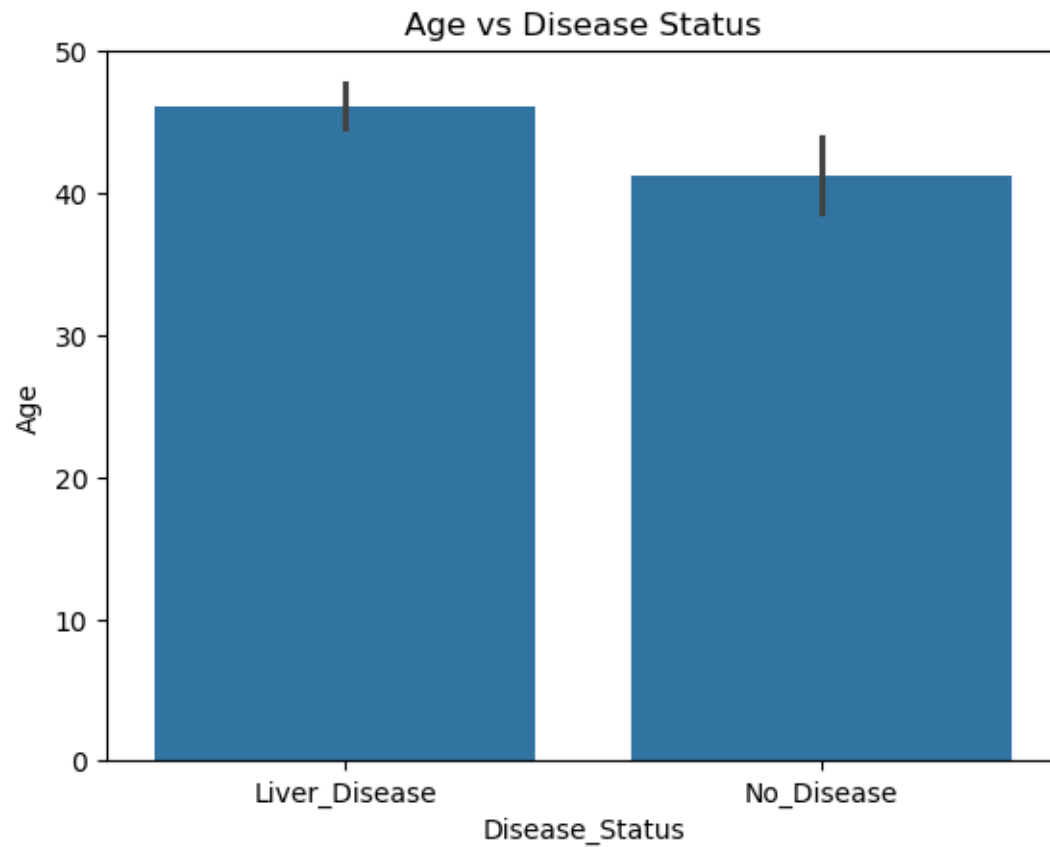
```
[94]: sns.boxplot(df , x='Disease_Status' , y='Age')  
plt.title('Age vs Disease Status')
```

```
[94]: Text(0.5, 1.0, 'Age vs Disease Status')
```



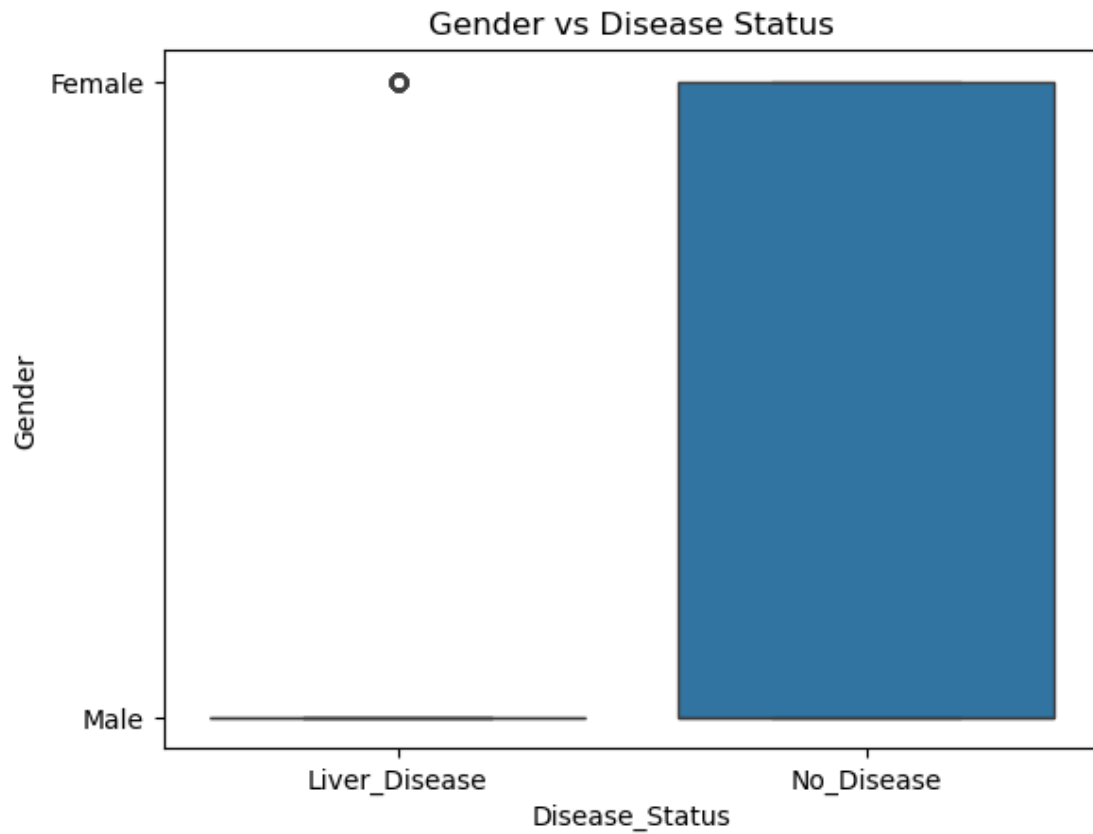
```
[71]: sns.barplot(df , x='Disease_Status' , y='Age')  
plt.title('Age vs Disease Status')
```

```
[71]: Text(0.5, 1.0, 'Age vs Disease Status')
```



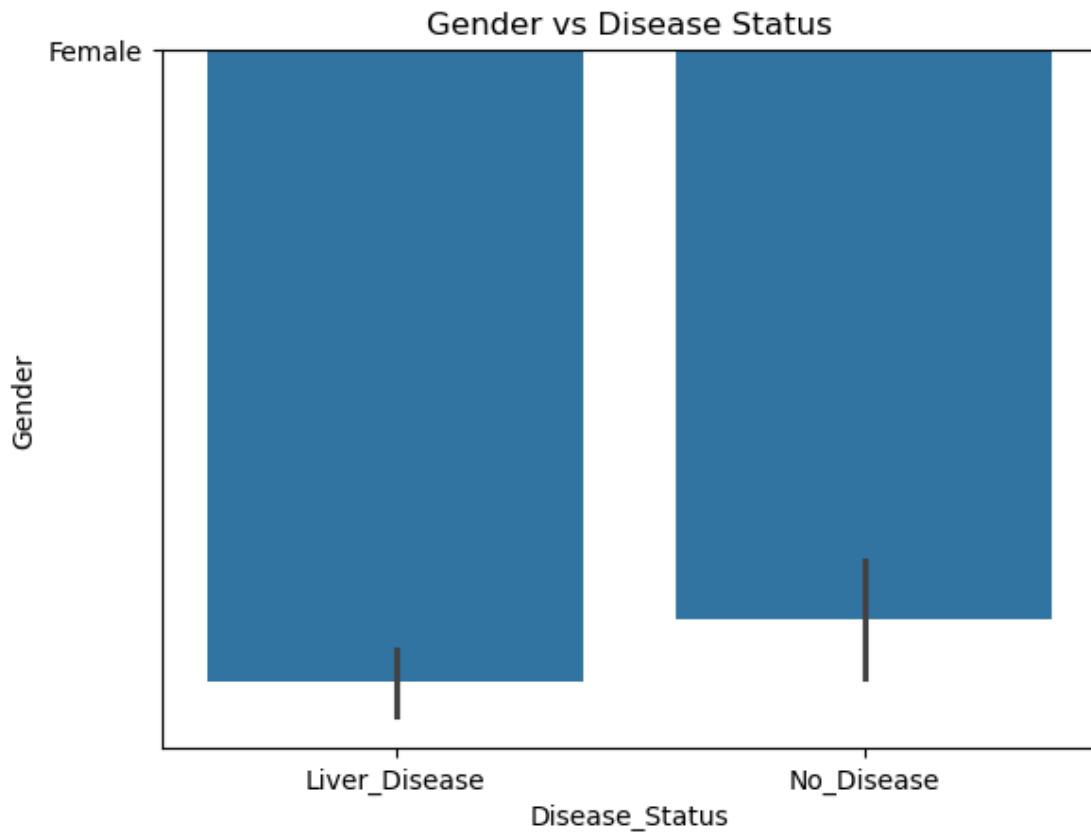
```
[95]: sns.boxplot(df , x= "Disease_Status" , y="Gender")  
plt.title('Gender vs Disease Status')
```

```
[95]: Text(0.5, 1.0, 'Gender vs Disease Status')
```



```
[73]: sns.barplot(df , x= "Disease_Status" , y="Gender")  
plt.title('Gender vs Disease Status')
```

```
[73]: Text(0.5, 1.0, 'Gender vs Disease Status')
```



```
[74]: df.groupby('Disease_Status')['Age'].mean()
```

```
[74]: Disease_Status
Liver_Disease    46.153846
No_Disease       41.239521
Name: Age, dtype: float64
```

```
[75]: df.groupby('Gender')['Disease_Status'].value_counts()
```

```
[75]: Gender  Disease_Status
Female  Liver_Disease      92
        No_Disease        50
Male    Liver_Disease     324
        No_Disease     117
Name: count, dtype: int64
```

```
[79]: df['Disease_Status'].value_counts()
```

```
[79]: Disease_Status
Liver_Disease    416
```

```
No_Disease      167
Name: count, dtype: int64
```

1.3 Conclusion:

This Report shows that the Quantity of patients with liver disease is more than non liver disease patients . Liver disease patients are = 416 and Non disease patients are = 167 . In Females There are 92 liver disease and 50 non liver disease . In Male there are 324 Liver disease and 117 non liver disease patients.